QIM HW6

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1 QIM Homework 6

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2 Question 1

We came up with two seperate strategies as guided by our conversations with both Professor Rothman and Damon Peterson.

Strategy A: Long/Short Offset Strategy (Guaranteed Loss via Costs) Strategy overview. Our strategy is to buy one unit of a stock and short one unit of the same stock at time $t = 0$. At $t = 9$ months, we liquidate both positions. This strategy is guaranteed to lose money because we pay a rebate rate on the short position over time, and we also incur transaction costs. The long and short positions offset each other, so there is no price risk.

Specific investment process. Say AAPL has price p_t at time t. At time $t = 0$, we purchase one share of AAPL and short one share of AAPL. For the long position, we incur transaction costs c_t. For the short position, assume the rebate rate is r_t, which may vary over time. We also must post margin equal to a fraction m of the position value.

At $t = 0$, the cash flow is

$$CF_0 = -c_t - m \cdot p_0$$

because the $-p_0$ from buying and the $+p_0$ from shorting cancel out, leaving only transaction costs and margin posted.

At time t, our cash flow is

$$CF_t = -r_t \cdot p_t$$

because we must continuously pay the daily borrowing/rebate cost on the short position.

At $t = 9$ months, our cash flow is

$$CF_9 = +m \cdot p_0$$

because we receive our margin back. We close the short position by delivering the share we already own from the long position. We have no cash flow from price movements of the underlying stock beause the gain/loss from our long will cancel the loss/gain from our short.

Since the long and short positions exactly offset, all price movements cancel out. The only remaining cash flows are transaction costs and borrowing costs, which are negative. Therefore, total profit is strictly negative, and the strategy is guaranteed to lose money.

Assumptions and Risks. We assume margin is posted in cash and returned at the end. We also assume that you have enough cash to meet any margin calls. Even if the stock price changes significantly, the long and short positions offset, so price movements do not affect the final outcome. Margin calls may affect timing of cash flows but do not change the fact that total profit is negative.

Note that even if you lend out your long AAPL position through the broker as well, you still lose money since you will pay the broker's fee plus lender's fee, and receive only the lender's fee. So net, you are negative.

Alternative strategy with "economic" exposure. The above strategy does not have "economic" exposure because you are going long and short the same stock. If you wanted an exposure to a stock and a guaranteed loss, then your strategy can be the following.

Keep trading every second a highly illiquid stock. Specifically, you should continue to buy and sell, keeping on crossing the bid-ask spread. If you do this at a high enough frequency over the 9 month time period, you can get arbitrarily high transaction costs.

Strategy B: High-Frequency Trading to Incur Transaction Costs **Strategy overview**. This strategy repeatedly buys and sells a stock at very high frequency over the 9-month period, deliberately incurring transaction costs and bid-ask spread losses. By trading frequently enough, total costs become arbitrarily large, guaranteeing a loss. Over all these transactions, the gain in the stock price cannot be arbitrarily large, so we can always trade with a sufficiently high frequency to guarantee a loss.

Assumptions and Risks.
We assume: - The bid-ask spread remains positive - We can execute trades continuously

3 Question 2

Strategy A (Flipped Long/Short) **Answer**. No, flipping the sign does not produce a guaranteed gain.

Explanation.
The flipped strategy is still: - long 1 AAPL and short 1 AAPL at $t = 0$
- close both at $t = 9$ months

This is the *same position* as before, since long and short cancel in terms of price exposure. You still: - pay transaction costs - pay short borrowing (rebate) costs

So all cash flows remain negative:

$$\text{Total P\&L} = -(\text{transaction costs}) - (\text{borrow costs}) < 0$$

Thus, flipping the sign does **not** change the outcome — you still have a guaranteed loss.

Note: A guaranteed gain would not be a problem for no-arbitrage, so long as the return does not exceed the risk-free rate. The rebate rate for GC stocks is usually set at the Fed Funds rate. If you took the position of the prime broker, you would be earning risk-free money at the risk-free rate, which does not violate no-arbitrage.

Strategy B (Flipped High-Frequency Trading) **Answer**. No, flipping the sign does not produce a guaranteed gain.

Explanation.
Flipping the strategy would mean: - enter and exit a short position at as of a high frequency as possible.

This will not gurantee a gain. You will still lose money because as you enter the short you will have to sell the stock and then repurchase it, incurring transaction costs. If you can do this on an order of minutes you can generate enough transaction costs such that any gain from shorting will be overwhelmed by transaction costs.

These frictions break symmetry, so a guaranteed loss does **not** become a guaranteed gain.

3.0.1 Sources

[1] QIM Lecture 6: Short Selling